Ethan Wen

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EDUCATION

Princeton University, Princeton, NJ — Expected May 2029

B.S.E. in Electrical & Computer Engineering | Cumulative GPA: 3.82

Fall 2026 Coursework: Contemporary Logic Design (Verilog); Introduction to Programming Systems (C)

PROJECTS

Serial-Controlled VGA Character Display*, Verilog, Lattice iCE40-HX1K, Yosys, nextpnr-ice40, IceStorm*

- 40x30 character grid at 640x480/60 Hz over UART; 4-state parser FSM, per-pixel font ROM fetch, no framebuffer.
- 585/1280 logic cells, 6/16 block RAMs, closes at 108.5 MHz; per-module benches plus an end-to-end frame compare.

EchoAssist*, Swift, SwiftUI, CoreML, Apple Intelligence Foundation Models* — Jun 2026 - Jul 2026

- Live speaker-labeled accessibility captions generated fully on-device by two streaming CoreML models (Parakeet ASR, Sortformer diarization) synced on a shared audio clock, running ~0.6s behind real time.
- On-device summarization via Apple Intelligence Foundation Models and translation into six languages.
- Built with a three-person team and submitted to the App Store; review appeal pending.

EXPERIENCE

Software Engineer, **Relay** — Jun 2026 - Present

- Shipped the friendship, concerts, messaging, and gamification systems for a five-person Flutter music-sharing app.
- Authored engineering design documents covering Firestore data modeling, transaction safety, security rules, and abuse and scale analysis.

Everyone Can Code Chicago, Chicago, IL — Jun 2026 - Present

Software Engineer Intern, Sept 2026 - Present | iOS App Dev & ML Program, Jun 2026 - Jul 2026

- App Accelerator (one of 24 selected): building an iOS app on a small product team through App Store launch.
- Summer program: advanced to the second round of the Apple-judged showcase, top 12 of 56 teams.

COS 126 Undergraduate Course Assistant, **Princeton University** — Sept 2026 - Present

- Code reviews and debugging help for students in Princeton's introductory computer science course.

HoagieHelp Developer, **Hoagie Club** — Feb 2026 - Present

- Building frontend components in Next.js, React, TypeScript, and Tailwind for a student academic support platform, with contributions to Python backend services.

ACTIVITIES

Princeton University Robotics Club*, Princeton, NJ* — Sept 2025 - Present

Tinker Tech Lead, Aug 2026 - Present | Tinker Member, Sept 2025 - Aug 2026

- Setting project direction and managing the parts budget for Tinker, the club's hardware subteam; firmware, bring-up, and debugging work including vintage iPod refurbishment.

HONORS

Engineering Futures Award, **Chicago Engineers' Foundation** — $5,000 merit award, 2026

2nd Place, **IEEE SSCS Arduino Contest** — 2024

SKILLS

Hardware & Digital Design: Verilog; RTL design; Yosys / nextpnr / IceStorm; Icarus Verilog / GTKWave; Lattice iCE40; Arduino / ESP32
Programming: C++; Python; Swift; Dart; TypeScript / JavaScript
Frameworks & Tools: SwiftUI; CoreML; Flutter; Firebase / Firestore; Next.js / React; Git; Xcode; LaTeX
Spoken: English (native); Cantonese (native); Mandarin (conversational)